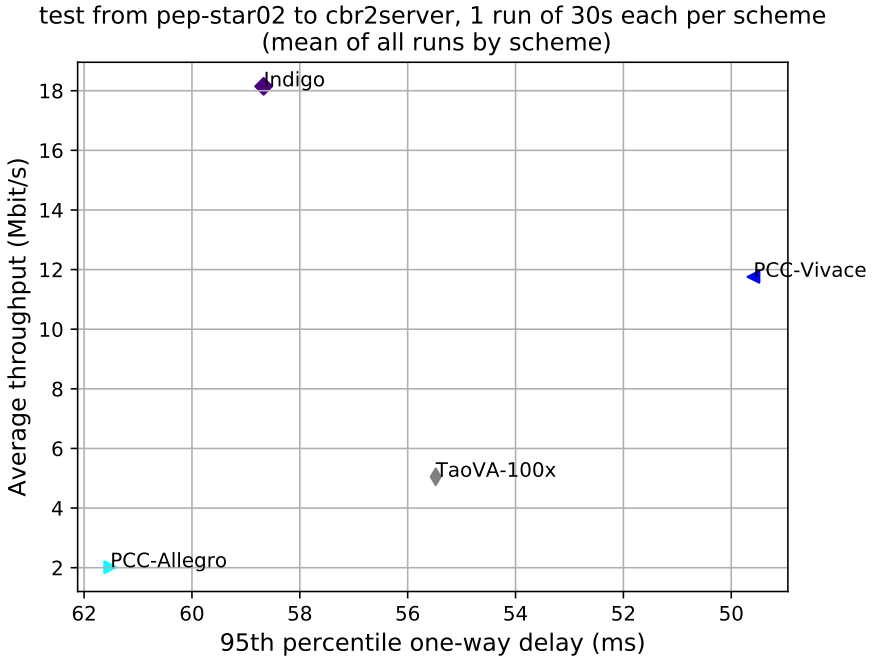
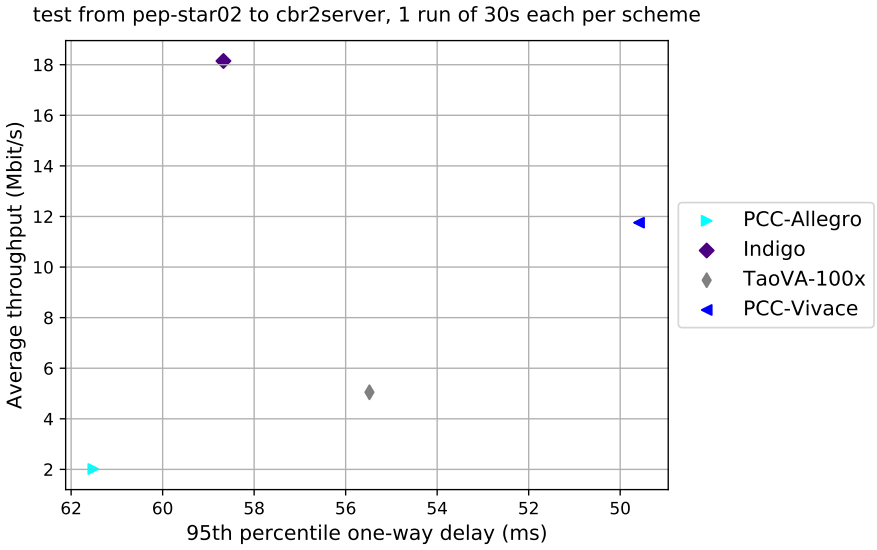
# Pantheon Report

Generated at 2026-11-09 23:39:49 (UTC).
Data path: pep-star02 (*local*) →cbr2server (*remote*).
Repeated the test of 4 congestion control schemes once.
Each test lasted for 30 seconds running 1 flow.
NTP offsets were measured against 202.65.114.202 and have been applied to correct the timestamps in logs.

```
System info:
Linux 7.0.0-42-generic
net.core.default_qdisc = fq
net.core.rmem_default = 212992
net.core.rmem_max = 4194304
net.core.wmem_default = 212992
net.core.wmem_max = 4194304
net.ipv4.tcp_rmem = 4096 131072 33554432
net.ipv4.tcp_wmem = 4096 16384 4194304

Git summary:
branch: HEAD @ 23e738ce5acae1d36e321886cd613b0b9401ac11
third_party/genericCC @ d0153f8e594aa89e93b032143cedbdfe58e562f4
third_party/indigo @ 463d89b09699a57bfdfbae351646df6a60040b90
third_party/pantheon-tunnel @ f866d3f58d27afd942717625ee3a354cc2e802bd
third_party/pcc @ 1afc958fa0d66d18b623c091a55fec872b4981e1
 M receiver/src/buffer.h
 M receiver/src/core.cpp
 M sender/src/buffer.h
 M sender/src/core.cpp
third_party/vivace @ 2baf86211435ae071a32f96b7d8c504587f5d7f4
```


test from pep-star02 to cbr2server, 1 run of 30s each per scheme
(mean of all runs by scheme)
Indigo
PCC-Vivace
TaoVA-100x
PCC-Allegro
Average throughput (Mbit/s)
95th percentile one-way delay (ms)
18
16
14
12
10
8
6
4
2
62
60
58
56
54
52
50


test from pep-star02 to cbr2server, 1 run of 30s each per scheme
Average throughput (Mbit/s)
95th percentile one-way delay (ms)
PCC-Allegro
Indigo
TaoVA-100x
PCC-Vivace
18
16
14
12
10
8
6
4
2
62
60
58
56
54
52
50

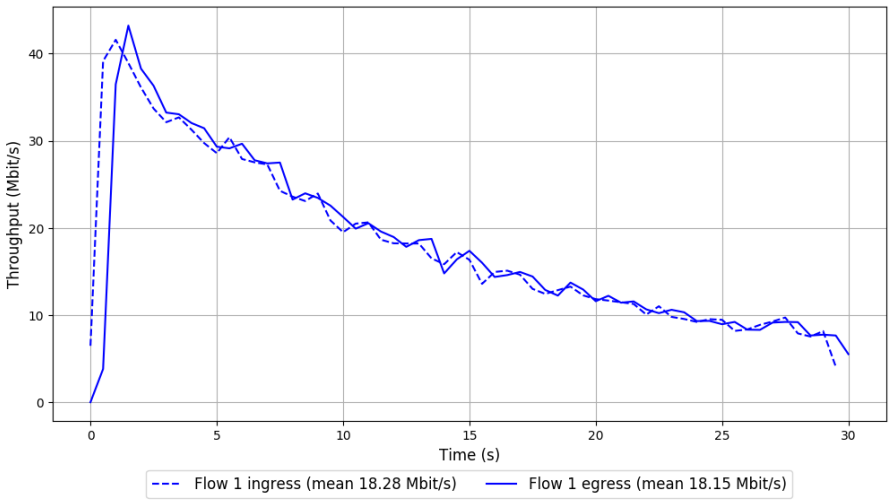
| scheme | # runs | mean avg tput (Mbit/s) flow 1 | mean 95th-%ile delay (ms) flow 1 | mean loss rate (%) flow 1 |
|---|---|---|---|---|
| Indigo | 1 | 18.15 | 58.67 | 0.73 |
| PCC-Allegro | 1 | 2.02 | 61.52 | 0.00 |
| TaoVA-100x | 1 | 5.05 | 55.48 | 0.77 |
| PCC-Vivace | 1 | 11.76 | 49.59 | 0.40 |

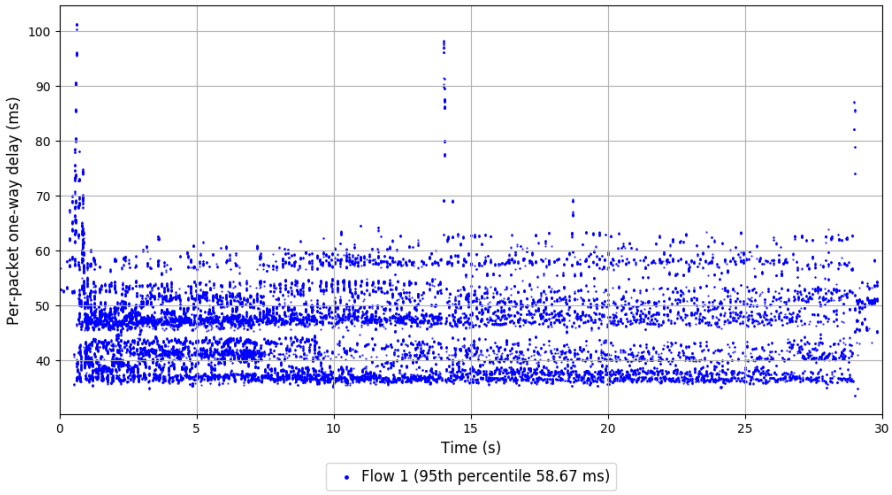
```
Run 1: Statistics of Indigo

Start at: 2026-11-09 22:28:42
End at: 2026-11-09 22:29:12
Local clock offset: 0.793 ms
Remote clock offset: -0.048 ms

# Below is generated by plot.py at 2026-11-09 23:39:48
# Datalink statistics
-- Total of 1 flow:
Average throughput: 18.15 Mbit/s
95th percentile per-packet one-way delay: 58.671 ms
Loss rate: 0.73%
-- Flow 1:
Average throughput: 18.15 Mbit/s
95th percentile per-packet one-way delay: 58.671 ms
Loss rate: 0.73%
```

Run 1: Report of Indigo — Data Link



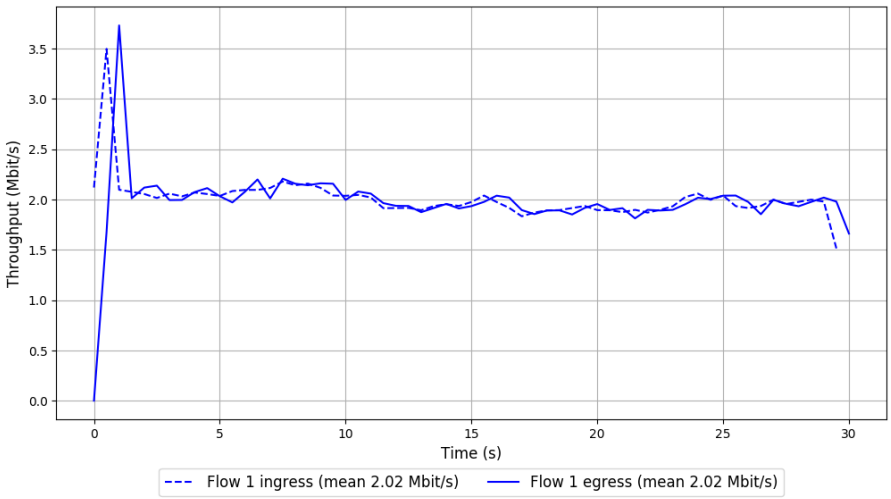



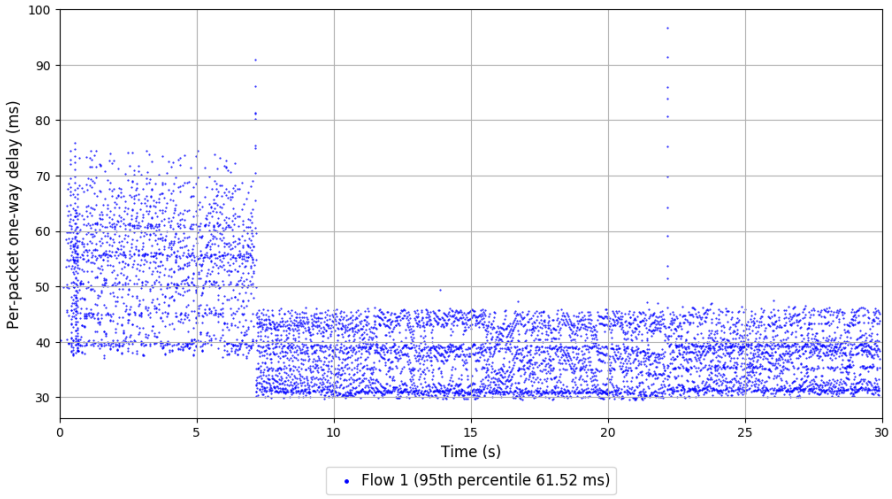
```
Run 1: Statistics of PCC-Allegro

Start at: 2026-11-09 22:27:34
End at: 2026-11-09 22:28:04
Local clock offset: 1.832 ms
Remote clock offset: 0.927 ms

# Below is generated by plot.py at 2026-11-09 23:39:48
# Datalink statistics
-- Total of 1 flow:
Average throughput: 2.02 Mbit/s
95th percentile per-packet one-way delay: 61.515 ms
Loss rate: 0.00%
-- Flow 1:
Average throughput: 2.02 Mbit/s
95th percentile per-packet one-way delay: 61.515 ms
Loss rate: 0.00%
```

Run 1: Report of PCC-Allegro — Data Link



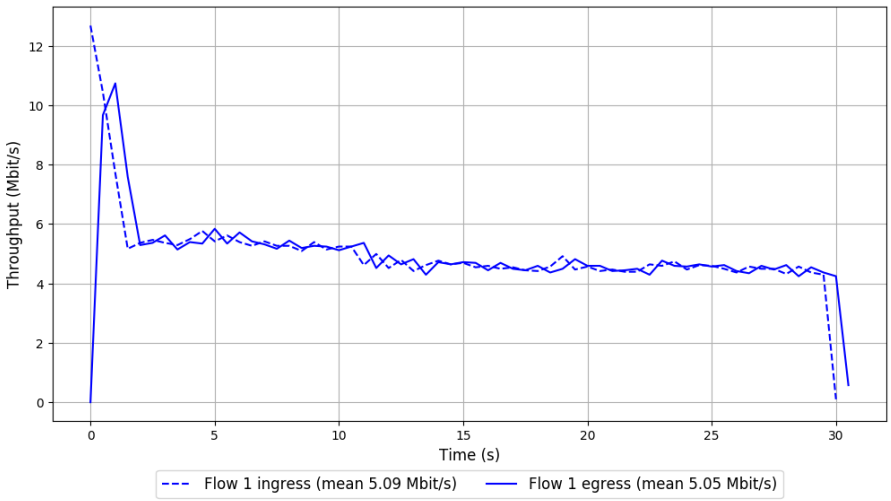



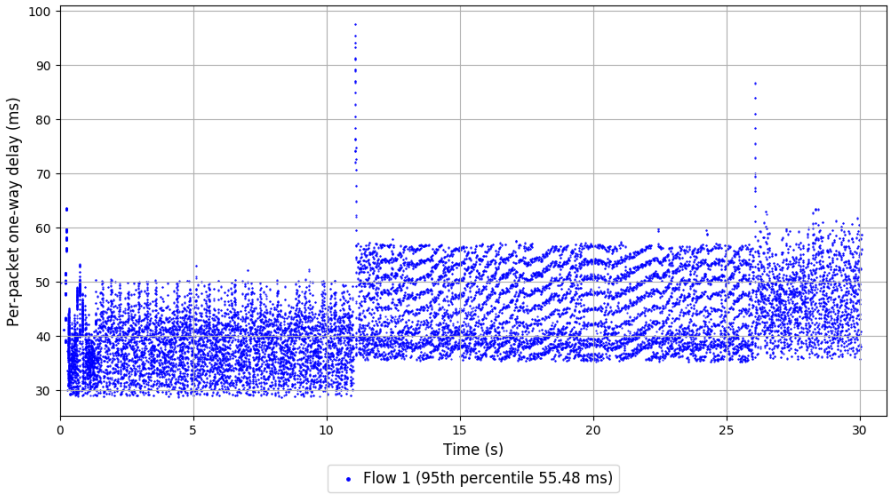
```
Run 1: Statistics of TaoVA-100x

Start at: 2026-11-09 22:25:15
End at: 2026-11-09 22:25:46
Local clock offset: 1.862 ms
Remote clock offset: -0.23 ms

# Below is generated by plot.py at 2026-11-09 23:39:48
# Datalink statistics
-- Total of 1 flow:
Average throughput: 5.05 Mbit/s
95th percentile per-packet one-way delay: 55.480 ms
Loss rate: 0.77%
-- Flow 1:
Average throughput: 5.05 Mbit/s
95th percentile per-packet one-way delay: 55.480 ms
Loss rate: 0.77%
```

Run 1: Report of TaoVA-100x — Data Link



```
Run 1: Statistics of PCC-Vivace

Start at: 2026-11-09 22:26:25
End at: 2026-11-09 22:26:55
Local clock offset: 1.764 ms
Remote clock offset: 0.845 ms

# Below is generated by plot.py at 2026-11-09 23:39:48
# Datalink statistics
-- Total of 1 flow:
Average throughput: 11.76 Mbit/s
95th percentile per-packet one-way delay: 49.589 ms
Loss rate: 0.40%
-- Flow 1:
Average throughput: 11.76 Mbit/s
95th percentile per-packet one-way delay: 49.589 ms
Loss rate: 0.40%
```

Run 1: Report of PCC-Vivace — Data Link

